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Counting Problems

1. 5 unique letters u, n, s, a, l

$5! = \boxed{120 \text{ unique subsets}}$

2. $\binom{13}{2} \cdot \binom{4}{2} \cdot \binom{4}{2} \cdot (44)$

$= \frac{13!}{2!\,11!} \cdot \frac{4!}{2!\,2!} \cdot \frac{4!}{2!\,2!} \times 44$

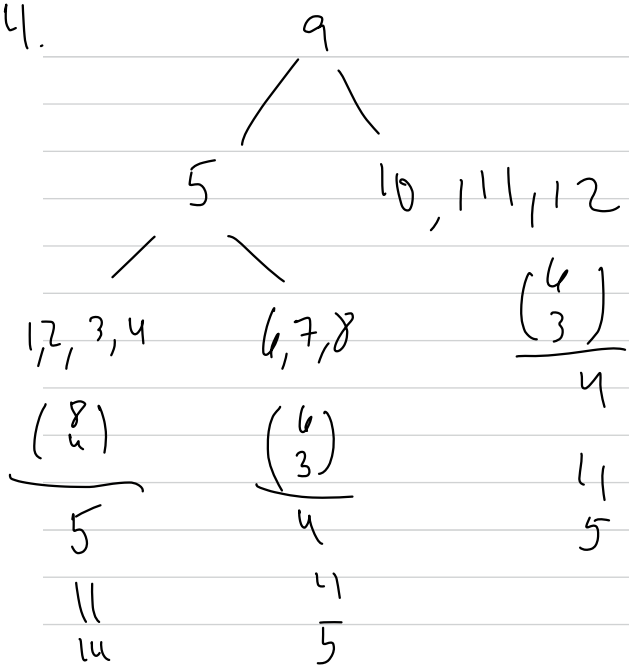
$= 78 \cdot 6 \cdot 6 \cdot 44 = \boxed{123,552}$

3. $\binom{16}{1} \cdot \binom{15}{6}$

$= \frac{16!}{1!\,15!} \cdot \frac{15!}{6!\,9!} = 16 \cdot 5005$

$= \boxed{80,080}$

4.



$14 \cdot 5 \cdot 5 = \boxed{350 \text{ trees}}$

5. 10 patients C time slot $\binom{10}{1} = 10$

4 nurses C shots $\binom{4}{1} = 4$

1 Nurse may or may not be on break

$$\binom{10}{4} = \frac{\cancel{10!}}{4!\,\cancel{6!}} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{5040}{24}$$

$= 210$ minutes or $3\frac{1}{2}$ hours

How many combinations are there for the # of patients served by the nurses

$$_{10}P_4 = \frac{10!}{\cancel{6!}} = 10 \cdot 9 \cdot 8 \cdot 7$$

$= \boxed{5040 \text{ combinations}}$